INCH-POUND

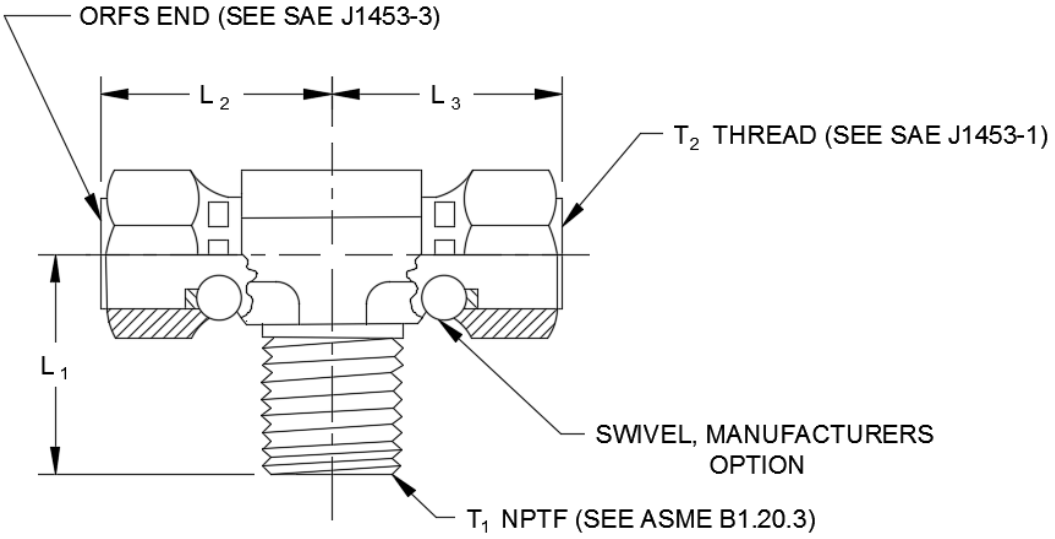
MIL-DTL-32372/10B
23 May 2019
SUPERSEDING
MIL-DTL-32372/10A
7 February 2017

DETAIL SPECIFICATION SHEET

# FITTING, ADAPTER, TEE, HYDRAULIC AND PNEUMATIC, ORFS FEMALE RUN TO RUN TO NPTF MALE ON BRANCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32372.




| Dash size | Tube OD nom. inches (mm) | T1 NPTF ASME B1.20.3 | T2 ORFS SAE J1453-1 |
|---|---|---|---|
| -4 | .250 (6.35) | 1/4 - 18 | 9/16 - 18 |
| -6 | .375 (9.53) | 1/4 - 18 | 11/16 - 16 |
| -8 | .500 (12.7) | 1/2 - 14 | 13/16 - 16 |
| -10 | .625 (15.88) | 1/2 - 14 | 1 - 14 |
| -12 | .750 (19.05) | 3/4 - 14 | 1 3/16 - 12 |
| -16 | 1.000 (25.40) | 1 7/16 - 12 | 1 7/16 - 12 |
| -20 | 1.250 (31.75) | 1 11/16 - 12 | 1 11/16 - 12 |
| -24 | 1.500 (38.10) | 1 1/2 - 11 1/2 | 2 - 12 |

See notes at end of figure.

FIGURE 1. Adapter, ORFS female run to run to NPTF male on branch.

AMSC N/A

FSC 4730

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| Dash size | D ID inches (mm) | L1 ± .030 (0.8) inches (mm) | L2 ±.030 0.8) Inches (mm) | L3 .030 Inches (mm) |
|---|---|---|---|---|
| -4 | .172 ±.003 (4.4 ±0.08)) | .780 (20) | 1.04 (26.5) | 1.04 (26.5) |
| -6 | .297 ±.004 (7.5 ±0.10) | 1.09 (28) | 1.14 (29.0) | 1.14 (29.0) |
| -8 | .391 ±.004 (9.9 ±0.10) | 1.22 (31) | 1.5 (38.0) | 1.5 (38.0) |
| -10 | .484 ±.004 (12.3 ±0.10) | 1.47 (37) | 1.61 (41.0) | 1.61 (41.0) |
| -12 | .609 ±.005 (15.5 ±0.13) | 1.59 (40) | 1.83 (46.5) | 1.83 (46.5) |
| -16 | .844 +.007/-.005 (21.4 ±0.13) | 1.97 (50) | 2.11 (53.5) | 2.11 (53.5) |
| -20 | 1.078 +.008/-.005 (27.4 +0.20/-0.13) | 2.38 (60) | 2.28 (58.0) | 2.28 (58.0) |
| -24 | 1.312 +.010/-.005 (33.3 ±0.25/-0.10) | 2.64 (67) | 2.4 (61.0) | 2.4 (61.0) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break all sharp edges and remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with ASME B1.20.3 for NPTF end and SAE J1453-1 for ORFS end.
5. For dimensions not shown, see SAE J1453-1, SAE J1453-3, or ASME B1.20.3.

FIGURE 1. Adapter, ORFS female run to run to NPTF male on branch - Continued.

REQUIREMENTS

Fitting configuration and dimensions shall be as specified on figure 1. Materials and finishes shall be in accordance with MIL-DTL-32372.

Finish. Finishes shall be in accordance with table I.

All plating's shall be capable of meeting 96-hour salt spray test in accordance with ASTM B117. The fittings shall show no evidence of red corrosion after 96 hours of salt spray. Fluid passages, other openings, and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

Maximum operating pressure. Maximum operating pressure shall be in accordance with MIL-DTL-32372.



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TABLE I. Material and finish identification codes.

| PIN code material/plating finish | Material | Plating finish |
|---|---|---|
| Blank | Steel | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 1/ 6/ |
| H | Steel | Zinc-aluminum in accordance with ASTM F1136/F1136M, grade 3, NC. |
| J | Steel | Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. 2/ |
| M | Nickel-copper alloy UNS N04400 | No additional finish. |
| N | High-chromium nickel alloy UNS N06690 | No additional finish. |
| P | Steel | Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class 1. 3/ |
| Q | Steel | Zinc plating in accordance with ASTM B633, type III, Fe/Zn 8. 3/ 4/ |
| R | Steel | Zinc plating in accordance with ASTM B633, type V, Fe/Zn 8. 3/ |
| S | Corrosion resistant steel | No additional finish. Passivation in accordance with SAE-AMS2700, method 1, type 6 or 7. |
| T | Titanium | Anodize in accordance with SAE-AMS2488 type 2. 5/ 7/ |
| TF | Titanium | Fluoride phosphate in accordance with SAE-AMS2486. 5/ 7/ |
| Z | Steel | Zinc plating in accordance with ASTM B695, type II, class 8. 3/ 4/ |
| ZC | Steel | Zinc may be any zinc plating from PIN codes J, Q, and Z with a colored chromate coating. 3/ 4/ |

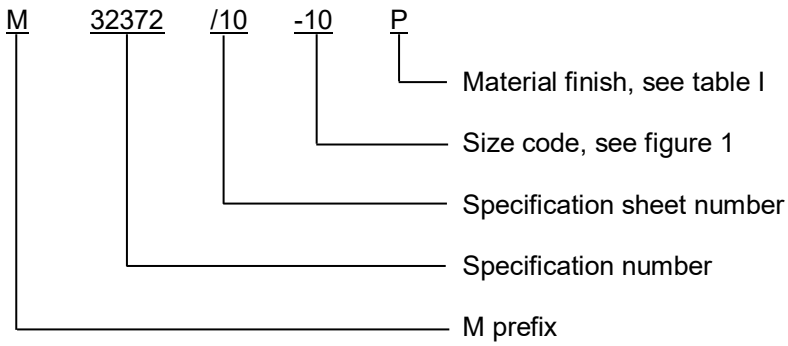
1/ Embrittlement test need not be run.
2/ The zinc-nickel alloy plate shall contain 12% to 16% nickel. The coating thickness shall be 315µ inches (8µm) minimum.
3/ Not for use in aircraft. Requires approval from the Program Office for all applications. Zinc-plated materials should be avoided in hydraulic systems with water-glycol fluids. Reference ASTM MNL37 - Fuels and Lubricants Handbook.
4/ These finishes have a chromate conversion coating or chromic acid rinse which includes hexavalent chromium and are not recommended.
5/ A pretreatment, a modification of the fluoride treatment, or a post treatment shall be applied so the final color of the fittings shall be similar to SAE-AMS-STD-595 colors 36076 through 36293
6/ Cadmium shall not be used in oxygen or potable water systems. Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.
7/ Titanium shall not be used in oxygen or potable water systems.

Further information on materials and finishes can be found in MIL-DTL-32372.



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PIN: The PIN consists of the letter M, the basic specification number, the specification sheet number, a number for fitting size, and a letter for the material finish designator.




PIN example: M32372/10-10P indicates a tee, fitting, .625 inch tube OD, steel with zinc phosphate finish.

O-ring is not supplied. Order separately, in accordance with SAE-J515.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Cadmium is prohibited from use in NAVSEA-owned shipboard systems in accordance with the requirements of T9070-AL-DPC-020/077-2, unless otherwise approved by NAVSEA.

Class I and II ozone depleting substances (ODS) shall not be used in MIL-DTL-32372/10 or any referenced procedures.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-32372, this document references the following:

| | | |
|---|---|---|
| MIL-DTL-16232 | SAE-AMS-STD-595/36099 | SAE-AMS-STD-595/36293 |
| ASME B1.20.3 | SAE-AMS-STD-595/36118 | SAE-AMS2417 |
| ASTM B117 | SAE-AMS-STD-595/36134 | SAE-AMS2486 |
| ASTM B633 | SAE-AMS-STD-595/36152 | SAE-AMS2488 |
| ASTM B695 | SAE-AMS-STD-595/36170 | SAE-AMS2700 |
| ASTM F1136/F1136M | SAE-AMS-STD-595/36173 | SAE J1453-1 |
| ASTM MNL37 | SAE-AMS-STD-595/36176 | SAE J1453-3 |
| SAE-AMS-C-81562 | SAE-AMS-STD-595/36231 | SAE-J515 |
| SAE-AMS-QQ-P-416 | SAE-AMS-STD-595/36251 | T9070-AL-DPC-020/077-2 |
| SAE-AMS-STD-595/36076 | SAE-AMS-STD-595/36270 | |
| SAE-AMS-STD-595/36081 | SAE-AMS-STD-595/36280 | |



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CONCLUDING MATERIAL

Custodians:
Army - AR
Navy - SH
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2019-037)

Review activities:
Army - AT, GL
Navy - AS, SA, YD
Air Force- 11

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online data at https://assist.dla.mil.



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